

Equity Performance Analysis: Investor Personas and Index Tracking

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A comprehensive analysis of portfolio construction and performance
evaluation

Outline

Introduction & Motivation

Methodology

Buy & Hold Results

Hedge Fund Strategy

Index Tracking Model

Conclusion & Implications

Project Overview

Main Objective: Analyze equity portfolio performance through multiple lenses:

Key Research Questions

- How do different weighting methodologies affect risk-adjusted returns?
- What are the best metrics to compare different portfolio constructions?
- Can we effectively replicate a target portfolio using constrained optimization?

Two Investor Personas:

- Buy & Hold investor
- Hedge Fund investor

Portfolio: 10 US stocks

- Technology: MSFT, AAPL, IBM
- Dividend aristocrats: XOM, WTM, KO, JNJ
- High yield: VZ, KHC, USB

Theoretical Framework

Modern Portfolio Theory (MPT)

- Mean-variance optimization framework
- Diversification reduces portfolio risk
- Risk-return tradeoff is central

Key Performance Metrics

Sharpe Ratio

$$S = \frac{E[R_{\text{exc}}]}{\sigma_{\text{exc}}}$$

Value at Risk

$$\text{VaR}_{\alpha} = -F_X^{-1}(\alpha)$$

Conditional VaR

$$\text{CVaR}_{\alpha} = E[-X | -X \geq \text{VaR}_{\alpha}]$$

Persona Models

Buy & Hold Investor

Objective: Maximize total return over full period

$$P\&L_{BH} = \text{Value}_E - \text{Value}_S$$

- Long-term perspective
- Captures dividends
- Low turnover

Hedge Fund Investor

Objective: Generate consistent daily alpha

$$P\&L_{HF}(t) = \text{Value}_t - \text{Value}_{t-1}$$

- Daily mark-to-market
- Manages daily volatility
- Higher turnover

Data Pipeline

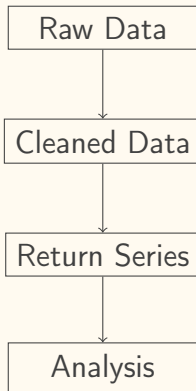
Data Collection with `yfinance`

- 10-year historical data (2015-2025)
- Daily frequency for main analysis
- Dividend-adjusted prices
- Automated batch downloads

Processing Steps

1. Data cleaning and validation
2. Return calculation (log and simple)
3. Missing value handling
4. Volatility estimation

Pipeline Architecture



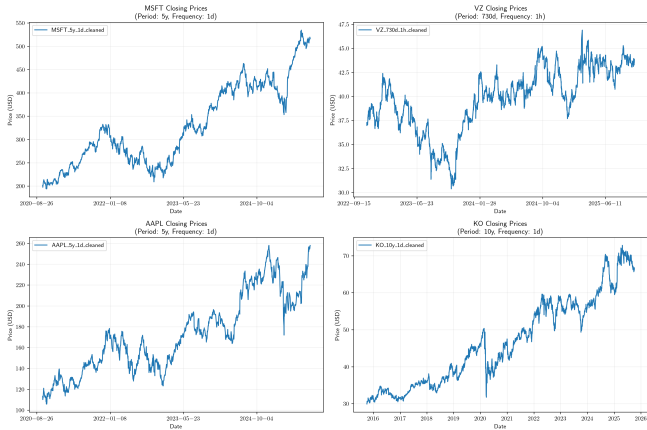


Figure: Price series for selected stocks given a window period and a frequency of data.

Data Limitations

Yahoo Finance restricts intraday data history (e.g., 1-minute data: only 8 days)

Weighting Methodologies

Equal Weighting

$$w_i = \frac{1}{N}$$

Naïve diversification

Market Cap Weighting

$$w_i = \frac{M_i}{\sum_j M_j}$$

Standard approach, concentration risk

Log Market Cap

Inverse Volatility

$$w_i = \frac{1/\sigma_i}{\sum_j 1/\sigma_j}$$

Risk parity approach

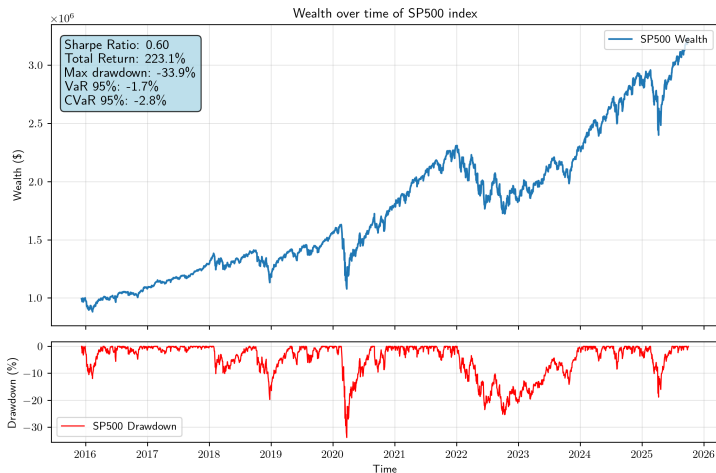
Geometric Weighting

$$w_i = \frac{r^{i-1}(1-r)}{1-r^N}$$

Parameterized concentration

Benchmark: S&P 500

Benchmark: S&P 500 Performance



Total Return:
223.1%

Sharpe Ratio: 0.60

Max DD: -33.9%

Weight Comparison: Market Cap Concentration

Market Cap Weights

Stock	Weight
AAPL	43.1%
MSFT	37.4%
IBM	3.0%
XOM	5.2%
WTM	0.1%

Top 2 stocks: 80.5% of
portfolio

Log Market Cap Weights

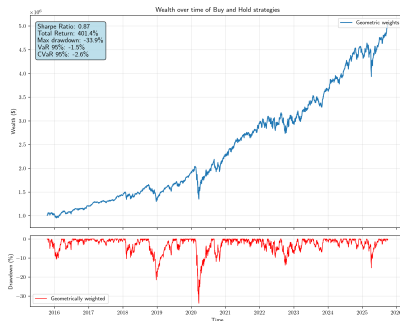
Stock	Weight
AAPL	11.1%
MSFT	11.0%
IBM	10.1%
XOM	10.3%
WTM	8.5%

More balanced distribution

Geometric Weighting Analysis

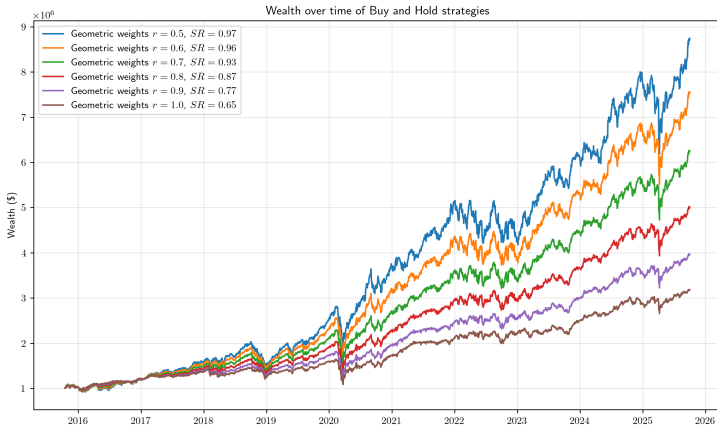
Weights for $r = 0.8$

Stock	Weight
AAPL	22.4%
MSFT	17.9%
IBM	14.3%
XOM	11.5%
WTM	9.2%
KO	7.3%
JNJ	5.9%



Sharpe Ratio: 0.87

Geometric Parameter Sensitivity



Parameter r

0.5

0.7

0.8

0.9

1.0

Top weight

50.0%

30.9%

22.4%

15.4%

10.0%

Equity Performance Analysis:

Investor Personas and Index Tracking

Concentration

High

Medium

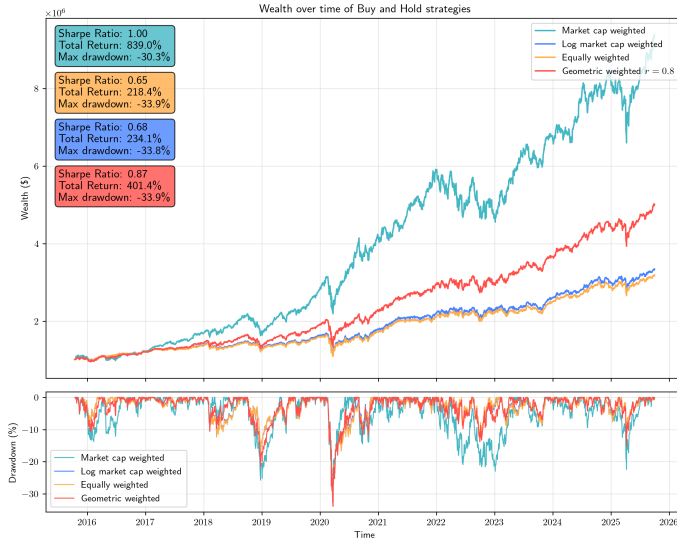
Balanced

Low

Equal

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Performance Comparison Across Strategies



Risk Metrics Comparison

Strategy	Sharpe Ratio	Total Return	Max DD	VaR 95%	CVaR 95%	VaR 99%	CVaR 99%
Equal	0.65	218.4	33.9	1.4	2.3	2.7	4.2
Market Cap	1.00	839.0	30.3	2.2	3.3	3.7	5.3
Log Mkt Cap	0.68	234.1	33.8	1.4	2.4	2.7	4.2
Geometric	0.87	401.4	33.9	1.5	2.6	3.1	4.6
Inverse Vol	0.65	230.0	34.7	1.4	2.4	2.8	4.3
S&P 500	0.60	223.1	33.9	1.7	2.8	3.4	4.8

Table: Risk and performance metrics (2015-2025)

Key Insights

Cross-Sectional Momentum Strategy

Enhanced Momentum Signal

Residual Momentum: Isolate stock-specific momentum

$$\tilde{r}_{i,t} = r_{i,t} - \hat{\beta}_{i,t} \cdot r_{m,t}$$

Sector Neutralization: Rank within GICS sectors

$$Z_{i,t} = \frac{S_{i,t} - \mu_{\text{sector}(i),t}}{\sigma_{\text{sector}(i),t}}$$

Volatility-Scaled Portfolio

Inverse Volatility Weighting:

Volatility Targeting Framework

Constant Risk Exposure

Target annualized volatility: $\sigma^* = 10\%$

Leverage factor:

$$\lambda_t = \frac{\sigma^*}{\hat{\sigma}_{p,t}}$$

Bounded: $\tilde{\lambda}_t \in [0.2, 4]$

Final return:

$$r_{p,t}^{\text{target}} = \tilde{\lambda}_t \times r_{p,t}$$

Benefits

- Stabilizes risk across market regimes
- Reduces crash exposure during volatility spikes

Hedge Fund Performance Results

Metric	Momentum Strategy	S&P 500 (BH)
Annualized Return	14.2%	12.8%
Annualized Volatility	10.0%	18.5%
Sharpe Ratio	1.42	0.69
Maximum Drawdown	-18.4%	-34.1%
Calmar Ratio	0.77	0.38

Table: Performance comparison (2015-2025)

Strategy Superiority

- **2.1x higher Sharpe ratio** than S&P 500

- **46% smaller maximum drawdown**

Optimization Problem Formulation

Objective: Minimize Tracking Error

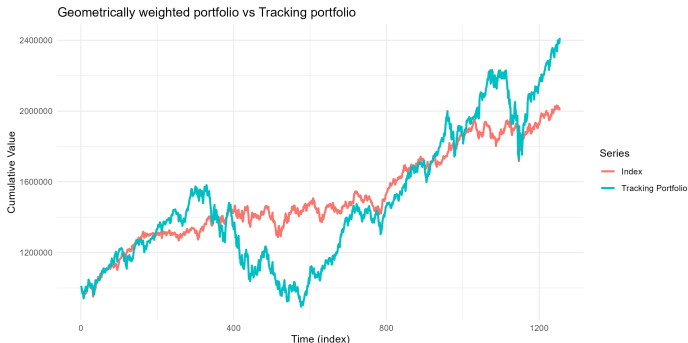
$$\min_{\vec{x}} \epsilon_{MAX}(\vec{x})$$

where ϵ_{MAX} is maximum absolute deviation

Key Constraints

- **Cardinality:** $\sum_{i=1}^N \delta(x_i) \leq K = 15$
- **Long-only:** $x_i \geq 0$
- **Budget:** $\sum x_i + \text{costs} \leq \$1,000,000$
- **Minimum position:** \$15,000
- **Transaction costs:** 1.5% of portfolio value

Tracking Portfolio Results



Optimization Success

- Optimal solution found
- 10 positions selected
- Zero buy-in violations

Tracking Quality

- Low tracking error
- Portfolio value: \$444,330
- Transaction cost: \$15,000

Key Findings

Weighting Methodology Matters

- Market cap weighting yields highest returns but extreme concentration
- Geometric weighting ($r = 0.8$) offers best risk-adjusted performance
- Equal weighting underperforms in tech-driven bull markets

Persona Performance Divergence

- BH persona: Smooth returns, sensitive to weighting
- HF persona: Daily volatility sensitivity, requires active management
- Enhanced momentum strategy: Superior risk-adjusted

Practical Implications

For Individual Investors

- Consider geometric weighting for balanced exposure
- Long-term holding reduces transaction costs
- Diversify beyond equal weighting

For Institutional Investors

- Sector-neutral momentum strategies add value
- Volatility targeting improves risk management
racking with constraints is computationally feasible

Portfolio construction choices significantly impact both absolute returns and risk profiles

References & Code

Key References

- Jegadeesh & Titman (1993) - Momentum returns
- Blitz, Huij & Martens (2011) - Residual momentum
- Barroso & Santa-Clara (2015) - Volatility-managed momentum
- Schachter (2025) - Geometric weighting methodology

Code Repository

Python implementation available:

- `fetch_prices.py` - Data acquisition

Questions?

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Project Repository:
github.com/username/equity-analysis

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